

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL-2

CSA0302-DATA STRUCTURES

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9.A real-time operating system uses a Red-Black tree to manage free memory blocks for its allocator, handling 100,000 allocation and deallocation requests per second. The developers chose Red-Black over AVL because it requires fewer rotations during insertions and deletions. However, when the allocation pattern alternates between allocating many small blocks and freeing them in random order, the tree becomes unbalanced in block size distribution, causing best-fit searches to traverse deep into the left subtree where small blocks reside. The team must consider augmenting the tree with subtree size information for order statistics or switching to a segregated free list for predictable $O(1)$ allocation.

Step 1: Represent Free Memory Blocks

Use **memory block size** as the key.

Enter:

10,50,5,30,89

Step1:

10	Insert		Delete		Find	Print	☐ Show Null Leaves
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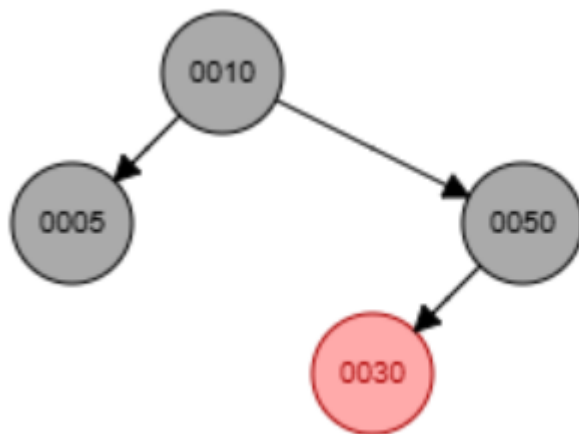
Step2:



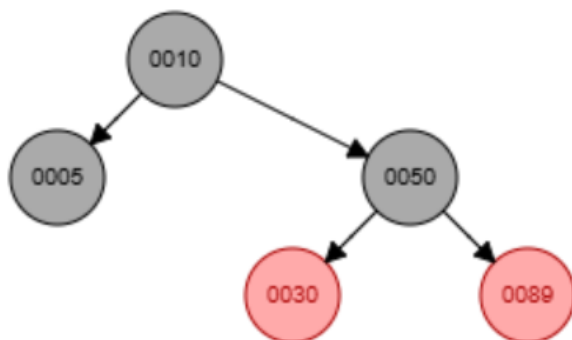
Step 3:



Step4:



Step6:



Conclusion:

The simulation demonstrates that a Red-Black Tree is suitable for managing dynamically changing free memory blocks because it maintains balanced search performance while reducing the number of rotations during frequent updates. For the given real-time workload, it provides efficient insertion, deletion and best-fit searching.

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